



Effects of dietary *Lycium barbarum* polysaccharide on the growth, antioxidant, immunity response, and intestinal microbiota in Bulatmai barbel (*Luciobarbus capito*)

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ABSTRACT

Chinese herbal polysaccharides has the functions of enhancing immunity, anti-oxidation, regulating intestinal flora, which was widely used in animals, but rarely used in fish. This study investigated the influence of dietary *Lycium barbarum* polysaccharide (LBP) on growth performance, digestive performance, immune performance, gene expression, and intestinal flora in Bulatmai barbel. The control cohort received a basal diet, whereas the treatment cohorts consumed identical basal diets comprising 0.10 %, 0.20 %, and 0.40 % LBP for 8 weeks. The findings indicated that compared with the control cohort, 0.10 % LBP notably enhanced the weight gain rate (WGR) and specific growth rate (SGR), while diminishing the feed coefficient (FCR). Overall, 0.10 % LBP markedly improved digestive performance and elevated trypsin, lipase, and amylase functional levels. Additionally, 0.10 % LBP markedly increased the antioxidant activity in Bulatmai barbel. LBP upregulates the mRNA expression of Heat shock proteins (HSP70), Alkaline phosphatase (AKP), and Superoxide dismutase (SOD). Of note, 0.10 % LBP can increase the diversity and richness of intestinal flora. Furthermore, LBP enhanced the proportional presence of fusobacteria while reducing proteobacteria levels. At the genus level, it elevated the proportional presence of *Cetobacterium*. Therefore, this study has shown that 0.10 % LBP positively regulated growth performance, digestive performance, immune performance, gene expression, and intestinal flora in Bulatmai barbel. LBP is a feed additive with excellent potential.

1. Introduction

Bulatmai barbel (*Luciobarbus capito*) belongs to the carp family Barbel and was imported into China from Uzbekistan (Geng et al., 2012; Zhang et al., 2023). It is a sizeable economic fish with saline–alkaline resistance and a fast growth rate in saline–alkaline aquaculture in China. In the early days of aquaculture, antibiotics or chemical drugs were used to reduce diseases in aquatic animals;

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however, excessive use has led to severe pollution of the microecological environment. Therefore, there is an urgent need to identify natural, nontoxic compounds as substitutes, and plant polysaccharides have become a research hotspot. Polysaccharides from aquaculture plants are widely used to promote growth, improve digestion and antioxidant properties, and increase immunity and disease resistance (Wu et al., 2019; Su et al., 2020; Yu et al., 2022).

Lycium barbarum L. belongs to the genus *Lycium* in the Nightshade family. *Lycium barbarum* L. is a perennial deciduous shrub with bright orange-red oval berries (Amagase et al., 2011). *Lycium barbarum* polysaccharide (LBP), serving as the principal bioactive component of *Lycium barbarum* L., exhibits a range of biological functions, including immune regulation, liver protection, vision improvement, nerve preservation, antioxidant and anti-aging properties, enhanced metabolic capacity, and potential benefits in countering cancer and diabetes (Zhang et al., 2014; Shah et al., 2018). Research has demonstrated that LBP exhibits considerable potential in enhancing the antioxidant capacity and immunity of aquatic animals, including *Trachinotus ovatus*, white-cloud mountain fish, and Nile tilapia, and promote their growth and development (Bai et al., 2011; Tan et al., 2019; Tao et al., 2022). In addition, LBP has protective and antioxidant effects against liver injury induced by carbon tetrachloride, and improves immunity (Liu et al., 2015). Therefore, we believe that LBP, as an immunomodulator, has a broad potential for use in aquaculture.

The fast cultivation of Bulatmai barbel benefits the exploitation and application of saline-alkaline water resources and improves the saline-alkaline water aquaculture industry in China (Sun et al., 2023; Luo et al., 2024). However, the effect of LBP on the Bulatmai barbel culture remains unclear. Therefore, we added different doses of LBP to the feed and fed Bulatmai barbel for 8 weeks to explore its effects on growth, serum biochemical parameters, digestive enzyme activity, immune enzyme activity, liver immune gene expression, and intestinal flora diversity and richness. These observations in this investigation can supply fundamental data for the LBP as a new feed additive in saline-alkaline water aquaculture.

2. Methodologies and materials

2.1. Experimental diets

As shown in Table 1, the control cohort's diet served as the base diet, incorporating protein sources from soybean meal, secondary meal, rapeseed meal, fish meal, chicken meal, and soybean protein concentrate, while utilizing fish oil and soybean oil as fat sources. The basal diets were supplemented with 0.00 %, 0.10 %, 0.20 %, or 0.40 % LBP. According to the corresponding feed formula, all ingredients were ground and blended in a mixer after sieving an 80-mesh screen. After mixing the vitamin premix, mineral premix, fish oil, and soybean oil in a stepwise ratio, diameter 3 mm pellet feed was prepared using a pelleting machine (Peixu Machinery Equipment Co., Ltd. Zhengzhou, China). It was dried, sealed, marked, and stored in a -20°C refrigerator.

Table 1

Formulation and chemical proximate composition of the experimental diets (% dry matter basis).

	Diets (LBP%)			
	Control	0.10 %	0.20 %	0.40 %
Ingredient (%)				
Soybean meal	34.50	34.50	34.50	34.50
Wheat middling	16.00	16.00	16.00	16.00
Rapeseed meal	15.00	15.00	15.00	15.00
Fish meal	8.00	8.00	8.00	8.00
Poultry by-product meal	11.00	11.00	11.00	11.00
Soy protein concentrate	3.00	3.00	3.00	3.00
Fish oil	2.75	2.75	2.75	2.75
Soybean oil	2.75	2.75	2.75	2.75
Sodium CMC	1.50	1.50	1.50	1.50
Vitamin premix ^a	0.60	0.60	0.60	0.60
Mineral premix ^b	0.40	0.40	0.40	0.40
Lysine	0.23	0.23	0.23	0.23
Allicin	0.05	0.05	0.05	0.05
Zelote power	0.45	0.45	0.45	0.45
Choline	0.50	0.50	0.50	0.50
Ca(H ₂ PO ₄) ₂	1.50	1.50	1.50	1.50
microcrystalline cellulose	1.77	1.67	1.57	1.37
<i>Lycium barbarum</i> polysaccharide ^c	0	0.10	0.20	0.40
Total	100.00	100.00	100.00	100.00
Proximate analysis				
Crude protein (%)	38.02	38.04	37.99	38.05
Crude lipid (%)	8.58	8.61	8.55	8.65
Crude ash (%)	7.45	7.50	7.52	7.48
Lysine	2.16	2.11	2.09	2.17
TP	1.47	1.51	1.46	1.42

^a see Luo et al., (2024) for further details.

^b see Luo et al., (2024) for further details.

^c *Lycium barbarum* polysaccharide was purchased from Shanxi Baiwei Biotechnology Co., Ltd. with a polysaccharide content of 51.2 %.

2.2. Experimental procedure

Bulatmai barbel was collected from Hulan Experimental Base of Heilongjiang Fisheries Research Institute, Chinese Academy of Fishery Sciences, with weight of (13.63 ± 1.48) g and body length of (12.38 ± 3.40) cm. Before the experiment, healthy Bulatmai barbel plants with the exact specifications were randomly selected to be the primary feed and acclimated for 7 days to acclimatize to the testing environment and setup before commencing the actual study. A sum of 360 uniformly-sized, healthy Bulatmai barbels were arbitrarily split into four experimental cohorts, featuring three replications per cohort, with 30 specimens in each replicate. The diets supplemented with 0.00 %, 0.10 %, 0.20 %, and 0.40 % LBP were fed once at 8:00, 13:00, and 18:00, respectively. The test water was tap water after aeration temperature control, the water temperature was (24.5 ± 0.5) °C, oxygen content exceeded 6.5 mg/L, pH 6.9–7.3, nitrite nitrogen and ammonia nitrogen remained below 0.01 mg/L. Daily water replacement occurred at 30 %, with constant aeration provided. The study lasted for 8 weeks. Throughout the experimental duration, daily observations included monitoring both feed consumption and survival rates of Bulatmai barbels.

2.3. Specimen acquisition

After concluding the breeding experiment, the Bulatmai barbels were starved for 24 h. Three Bulatmai barbels were arbitrarily picked from each cycle cylinder, and their weight, body length, and organ weight were ascertained after anesthesia with eugenol (Chengdu Aikeda Chemical Reagent Co., Ltd.). The blood was procured from the caudal vein of Bulatmai barbels with a 1.5 mL sterile centrifuge tube (1 % heparin sodium wash) on an ice tray. The blood was left to stay still for 24 h at 4 °C, succeeded by centrifugation at 3000 r/min for 15 min with a high-speed centrifuge at 4 °C. The supernatant was then drawn and placed in a 1.5 mL centrifuge tube and was marked for the detection of serum biochemical indices. It was preserved in an ultra-low temperature freezer at -80 °C until further examination. Subsequently, the fishes were dissected, the internal organs were removed and weighed, the liver and intestines were separated successively, and the liver was weighed. The intestines were then divided into 1.5 mL centrifugal tubes and marked and stored at -80 °C for use.

2.4. Growth performance

The following formula calculates the growth performance indexes :

$$\text{Weight gain rate (WGR, \%)} = [W_t - W_0]/W_0 \times 100\%$$

$$\text{Specific growth rate (SGR, \%}\cdot\text{d}^{-1}) = [\ln W_t - \ln W_0]/d \times 100\%$$

$$\text{Survival rate (SR, \%)} = N_t / N_0 \times 100\%$$

$$\text{Feed conversion ratio (FCR, \%)} = (T - S)/(W_t - W_0)$$

$$\text{Hepatosomatic index (HSI, \%)} = (W_h/W_t) \times 100\%$$

$$\text{Viscerosomatic index (VSI, \%)} = (W_v/W_t) \times 100\%$$

$$\text{Condition factor (CF, \%)} = W_t/L^3 \times 100\%$$

In the above formula, W_0 and W_t signify the initial and final body weights of the experimental fish, respectively. N_0 and N_t signify the mantissa of fish in the cage at the beginning of the culture experiment and the mantissa of fish in the cage at the end, respectively. T and S represent the total and residual feed amounts, respectively. W_h is the liver mass (g), W_v is the visceral mass (g), d is the number of feeding days, and L is the body length of the fish.

2.5. Serum biochemical parameters analysis

Serum physiological and biochemical parameters (see Luo et al., 2024 for further details) were measured utilizing HITACHI-7170A automatic biochemical analyzer (Tokyo, Japan).

2.6. Enzymes activities analysis

The intestinal samples underwent homogenization with physiological saline at a 1:9 ratio (weight to volume) followed by centrifugation at 3000 g (4 °C) for 10 min. The obtained supernatant fractions were utilized to assess enzymatic activity parameters. The measurement of enzymes activities (see Luo et al., 2024 for further details) were executed employing commercial assay kits from Nanjing Jiancheng Bioengineering Institute (Nanjing, China).

2.7. Quantitative real-time PCR

Total RNA in livers was procured from Trizol reagent and then reverse-transcribed into cDNA by PrimeScript™ RT Reagent kit (Takara, Japan). The quantitative real-time PCR (qRT-PCR) reaction system was 10 µL. The amounts of reagents added are listed in

Table S1. The *HSP70*, *AKP*, and *SOD* gene expression levels were quantified relative to the expression of *18S* using the optimized by $2^{-\Delta\Delta Ct}$ value method. The qRT-PCR primers used in this study are listed in Table S2.

2.8. High throughput sequencing and data analysis

Within the control cohort, the intestinal content specimens of 0.10 %, 0.20 %, and 0.40 % LBP were amplified with co-primers (515 F and 806 R) for the V3-V4 region of the 16S rRNA gene. Genomic DNA extraction was performed utilizing a TianGen soil and fecal kit through magnetic bead methodology. The NEBNext Ultra II FS DNA PCR-free Library Prep Kit (New England Biolabs) facilitated library construction. Qubit and Q-PCR were used to quantify the libraries. Following library validation, a NovaSeq 6000 PE 250 platform was used for on-machine sequencing. Software quality control was performed on the original sequencing sequence and splicing using FLASH (version 1.2.11, <http://ccb.jhu.edu/software/FLASH/>) and FASTP software (Version 0.23.1). OTU clustering of sequences with 97 % similarity and elimination of chimeras, species accumulation box map to evaluate community richness and sampling adequacy. The Venn Diagram function within R software generated corresponding visualizations. QIIME2 software facilitated Alpha diversity assessment through Chao1, Shannon, and Simpson indices. Beta diversity assessment employed Bray-Curtis' principal coordinate analysis (PCoA), implemented via R software's ade4 and ggplot2 packages. Heatmaps were created utilizing R software.

2.9. Ethical statement

All experimental animal procedures were executed per the Guide for the Care and Use of CAFS Laboratory Animals at the Heilongjiang River Fisheries Research Institute. The experimental protocols were evaluated and authorized by the Laboratory Animal Welfare and Ethics Committee of the Heilongjiang River Fisheries Research Institute. The project identification code was 20211001-001, and the date of approval was October 15, 2021.

2.10. Statistical analysis

Data are denoted as mean $\pm$ standard deviation and underwent statistical evaluation utilizing SPSS 17.0. One-way analysis of variance (ANOVA) and multiple comparisons (Duncan's method) were applied to test the mean differences among all the data (see Luo et al., 2024 for further details). $P < 0.05$ was considered statistically significant.

3. Results

3.1. Growth performance

The experimental cohorts received varying doses of LBP (0.00 %, 0.10 %, 0.20 %, and 0.40 %) to evaluate the Bulatmai barbels' growth performance, as depicted in Table 2. The data revealed that finish body weights (FBW), weight gain rate (WGR), and specific growth rate (SGR) exhibited an initial increase followed by a decline as dietary LBP levels rose. Notably, the 0.10 % LBP cohort yielded substantially higher final mass and weight gain rates compared to other treatments ($P < 0.05$). The WGR observed in the 0.10 % LBP cohort exceeded the control by 16.73 %, while the SGR in the 0.40 % LBP cohort was markedly lower than the control, 0.10 %, and 0.40 % LBP cohorts ($P < 0.05$). Moreover, no notable variations emerged between the control and the 0.10 % and 0.40 % LBP cohorts. The control cohort displayed markedly higher VSI values than the 0.10 % and 0.40 % LBP cohorts ($P < 0.05$), while the 0.20 % LBP cohort showed markedly elevated VSI versus the 0.10 % LBP cohort ($P < 0.05$). The analysis revealed no significant variations in initial body weights (IBW), survival rate (SR), feed conversion ratio (FCR), hepatosomatic index (HSI), viscerosomatic index (VSI), and

Table 2

Growth performance of Bulatmai barbel fed diets with diverse dose LBP supplementation after 8 weeks (Mean $\pm$ SE, $n = 3$).

Parameters	Diets (LBP%)				Pr > F ^a		
	0.00 %	0.10 %	0.20 %	0.40 %	AVONA	Linear trend	Quadratic trend
IBW (g)	13.64 $\pm$ 0.14	13.66 $\pm$ 0.09	13.61 $\pm$ 0.09	13.79 $\pm$ 0.11	0.00	0.01	0.01
FBW (g)	41.33 $\pm$ 2.93 ^{bc}	47.79 $\pm$ 1.18 ^a	42.68 $\pm$ 1.43 ^b	36.65 $\pm$ 3.67 ^{bc}	0.00	0.01	0.14
WGR (%)	214.11 $\pm$ 29.77 ^b	249.92 $\pm$ 6.68 ^a	213.59 $\pm$ 10.74 ^b	196.68 $\pm$ 5.04 ^b	0.04	0.04	0.10
SGR (%)	2.00 $\pm$ 0.17 ^a	2.23 $\pm$ 0.03 ^a	2.00 $\pm$ 0.07 ^a	1.76 $\pm$ 0.14 ^b	0.01	0.02	0.19
SR (%)	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0	0.00	0.00	0.00
FCR (%)	60.54 $\pm$ 7.06	49.75 $\pm$ 3.97	57.64 $\pm$ 7.07	61.05 $\pm$ 1.47	0.11	0.06	0.13
HSI (%)	1.91 $\pm$ 0.16	1.50 $\pm$ 0.04	1.71 $\pm$ 0.10	1.58 $\pm$ 0.06	0.00	0.01	0.01
VSI (%)	10.79 $\pm$ 0.18 ^a	10.15 $\pm$ 0.48 ^c	10.72 $\pm$ 0.15 ^{ab}	10.16 $\pm$ 0.47 ^{bc}	0.22	0.16	0.06
CF (%)	0.79 $\pm$ 0.01	0.79 $\pm$ 0.01	0.80 $\pm$ 0.04	0.76 $\pm$ 0.04	0.57	0.51	0.59

The experimental groups were fed with different concentrations of LBP (0.00 %, 0.10 %, 0.20 %, 0.40 %) to determine the growth performance of Bulatmai barbel. See Luo et al. (2024) for further details. A row of data in the table (mean $\pm$ standard deviation) without superscript letters or the same superscript letters indicates no significant difference ($P > 0.05$). Different means significant difference ($P < 0.05$).

^a Significance probability associated with the F-statistic.

condition factor (CF) across all treatments, with SR maintaining 100 % throughout.

3.2. Serum biochemical parameters analysis

The experimental cohorts received varying LBP doses (0.00 %, 0.10 %, 0.20 %, and 0.40 %) to determine the serum biochemical parameters in Bulatmai barbels in Table 3. As depicted in Table 3, with the increase in dietary LBP, triglyceride decreased at first and then increased, and triglyceride in the 0.10 % LBP cohort was markedly lower than that in the control cohort and 0.40 % LBP cohort ($P < 0.05$). Urea nitrogen level decreased with the increase in LBP supplemental level, and urea nitrogen level in the 0.40 % LBP cohort was markedly lower than that in the control cohort ($P < 0.05$). There were no significant differences in GLU, ALT, AST, ALP, ALB, GLO, CHOL, and CRE levels among all cohorts.

3.3. Digestive enzyme activities

The experimental cohorts received varying doses of LBP (0.00 %, 0.10 %, 0.20 %, and 0.40 %) to evaluate the digestive enzyme performance in Bulatmai barbels, as depicted in Table 4. The data reveals that trypsin levels in the 0.10 % LBP cohort exhibited notably greater values compared to the control, 0.20 %, and 0.40 % LBP cohorts ($P < 0.05$). The specimens exposed to 0.20 % and 0.40 % LBP demonstrated markedly elevated trypsin activity versus the control cohort ($P < 0.05$). The control cohort displayed substantially reduced lipase levels in contrast to the 0.10 %, 0.20 %, and 0.40 % LBP cohorts ($P < 0.05$). Moreover, no substantial variations in lipase activity were observed among the three LBP cohorts. The 0.10 % LBP cohort showed considerably enhanced amylase activity in comparison to the control, 0.20 %, and 0.40 % LBP cohorts ($P < 0.05$). Furthermore, no notable distinctions emerged between the control and the 0.20 % and 0.40 % LBP cohorts.

3.4. Immune enzyme activity

The experimental cohorts received varying doses of LBP (0.00 %, 0.10 %, 0.20 %, and 0.40 %) to evaluate immunoenzyme activity levels in Bulatmai barbels as depicted in Table 5. Analysis of Table 5 reveals that ACP levels in the control cohort exceeded those of the other experimental cohorts markedly ($P < 0.05$), while no notable variance emerged between the 0.10 % and 0.20 % LBP cohorts. AKP activity exhibited an initial rise followed by a decline as dietary LBP levels increased. The 0.10 % LBP cohort yielded substantially higher AKP activity compared to the control, 0.20 %, and 0.40 % LBP cohorts ($P < 0.05$), whereas control and 0.40 % LBP cohorts showed comparable levels.

The T-AOC, CAT, GSH-PX, Na^+/K^+ -ATPase, and MDA levels demonstrated an initial elevation followed by reduction as dietary LBP supplementation increased. The 0.10 % LBP cohort produced notably elevated T-AOC, CAT, GSH-PX, and Na^+/K^+ -ATPase levels versus

Table 3

Impact of varying LBP dietary concentrations on serum biochemical parameters of Bulatmai barbel (Mean $\pm$ SE, $n = 3$).

Parameters	Dietary LBP levels				Pr > F ^a		
	0.00 %	0.10 %	0.20 %	0.40 %	AVONA	Linear trend	Quadratic trend
TP(g/L)	37.77 $\pm$ 2.84	36.87 $\pm$ 1.24	37.33 $\pm$ 3.05	38.77 $\pm$ 5.06		0.84	0.96
GLU (mmol/L)	2.64 $\pm$ 0.16	2.46 $\pm$ 0.05	3.24 $\pm$ 0.86	2.67 $\pm$ 0.31	0.91	0.19	0.09
ALT (U/L)	9.67 $\pm$ 2.31	8.33 $\pm$ 3.06	11.33 $\pm$ 3.06	13.67 $\pm$ 2.89	0.27	0.46	0.52
AST (U/L)	265.67 $\pm$ 38.81	280.33 $\pm$ 24.01	306.33 $\pm$ 23.29	287.67 $\pm$ 11.59	0.20	0.43	0.43
TG (mmol/L)	4.86 $\pm$ 0.20 ^{ab}	4.21 $\pm$ 0.22 ^c	4.50 $\pm$ 0.23 ^{bc}	5.01 $\pm$ 0.13 ^a	0.36	0.00	0.19
ALP (U/L)	16.67 $\pm$ 2.89	17.33 $\pm$ 10.97	20.33 $\pm$ 7.64	16.00 $\pm$ 7.81	0.00	0.77	0.65
ALB (g/L)	15.63 $\pm$ 0.91	15.23 $\pm$ 0.90	15.47 $\pm$ 1.00	14.80 $\pm$ 0.53	0.91	0.76	0.51
GLO (g/L)	22.13 $\pm$ 1.99	21.63 $\pm$ 0.74	21.87 $\pm$ 2.12	23.97 $\pm$ 4.53	0.67	0.71	0.88
CHOL (mmol/L)	12.05 $\pm$ 1.15	11.46 $\pm$ 1.38	12.04 $\pm$ 1.95	11.45 $\pm$ 0.66	0.72	0.80	0.52
CRE (umol/L)	60.00 $\pm$ 26.51	67.00 $\pm$ 12.17	58.67 $\pm$ 24.83	48.33 $\pm$ 21.73	0.90	0.78	0.82
BUN (mmol/L)	1.71 $\pm$ 0.13 ^a	1.19 $\pm$ 0.43 ^{ab}	1.31 $\pm$ 0.28 ^{ab}	1.06 $\pm$ 0.26 ^b	0.78	0.34	0.21
					0.11		

The experimental groups were fed with different concentrations of LBP (0.00 %, 0.10 %, 0.20 %, 0.40 %) to determine the serum biochemical parameters of Bulatmai barbel. See Luo et al. (2024) for further details. Values are means of three replicate groups $\pm$ SD. Values without a common superscript within a row differ significantly ($P < 0.05$).

^a Significance probability associated with the F-statistic.

Table 4Digestive enzyme activities of Bulatmai barbel fed diets with diverse dose LBP supplementation after 8 weeks (Mean $\pm$ SE, n = 3).

Parameters	Diets (LBP%)				Pr > F ^a		
	0.00 %	0.10 %	0.20 %	0.40 %	AVONA	Linear trend	Quadratic trend
Trypsin(U/mg)	1056.76 $\pm$ 74.39 ^c	2511.75 $\pm$ 31.00 ^a	1970.15 $\pm$ 153.24 ^b	2134.11 $\pm$ 219.88 ^b	0.00	0.00	0.00
Lipase(U/g)	1.32 $\pm$ 0.12 ^b	2.28 $\pm$ 0.18 ^a	2.25 $\pm$ 0.09 ^a	2.40 $\pm$ 0.16 ^a	0.00	0.00	0.14
Amylase(U/mg)	14.27 $\pm$.92 ^b	19.16 $\pm$ 0.64 ^a	14.27 $\pm$ 0.19 ^b	14.59 $\pm$ 1.18 ^b	0.04	0.23	0.22

The experimental groups were fed with different concentrations of LBP (0.00 %, 0.10 %, 0.20 %, 0.40 %) to determine the digestive enzyme activities of Bulatmai barbel. Values are means of three replicate groups $\pm$ SD. Values without a common superscript within a row differ significantly ($P < 0.05$).

^a Significance probability associated with the F-statistic.

Table 5Immunoenzyme activity activities of Bulatmai barbel fed diets with different dose of LBP as supplement after 8 weeks (Mean $\pm$ SE, n = 3).

Parameters	Diets (LBP%)				Pr > F ^a		
	0.00 %	0.10 %	0.20 %	0.40 %	AVONA	Linear trend	Quadratic trend
ACP(U/g protein)	370.37 $\pm$ 16.61 ^a	310.40 $\pm$ 5.90 ^c	295.87 $\pm$ 3.88 ^c	329.56 $\pm$ 2.55 ^b	0.00	0.00	0.23
AKP (U/g protein)	105.91 $\pm$ 7.39 ^c	210.91 $\pm$ 6.47 ^a	177.23 $\pm$ 14.58 ^b	124.38 $\pm$ 15.86 ^c	0.00	0.00	0.00
T-AOC (U/mg protein)	1.40 $\pm$ 0.07 ^b	1.93 $\pm$ 0.18 ^a	1.53 $\pm$ 0.95 ^b	1.39 $\pm$ 0.14 ^b	0.00	0.00	0.01
SOD (U/mg protein)	1.61 $\pm$ 0.38 ^c	2.46 $\pm$ 0.13 ^b	2.47 $\pm$ 0.30 ^b	3.50 $\pm$ 0.44 ^a	0.00	0.15	0.06
CAT (U/mg protein)	37.74 $\pm$ 6.03 ^b	44.71 $\pm$ 1.06 ^a	34.31 $\pm$ 0.84 ^b	32.59 $\pm$ 0.94 ^b	0.01	0.01	0.01
GSH-PX (U/mg protein)	69.23 $\pm$ 8.12 ^b	103.22 $\pm$ 15.13 ^a	69.15 $\pm$ 1.78 ^b	68.07 $\pm$ 7.35 ^b	0.00	0.00	0.00
Na ⁺ K ⁺ ATP (U/mg protein)	0.23 $\pm$ 0.02 ^b	0.47 $\pm$ 0.03 ^a	0.26 $\pm$ 0.02 ^b	0.23 $\pm$ 0.02 ^b	0.00	0.00	0.00
MDA (nmol/mg protein)	0.48 $\pm$ 0.04 ^a	0.50 $\pm$ 0.05 ^a	0.34 $\pm$ 0.02 ^b	0.27 $\pm$ 0.04 ^b	0.00	0.03	0.03

The experimental groups were fed with different concentrations of LBP (0.00 %, 0.10 %, 0.20 %, 0.40 %) to determine the immunoenzyme activity activities of Bulatmai barbel. See Luo et al. (2024) for further details. Values are means of three replicate groups $\pm$ SD. Values without a common superscript within a row differ significantly ($P < 0.05$).

^a Significance probability associated with the F-statistic.

other treatments ($P < 0.05$). MDA measurements in the 0.20 % and 0.40 % LBP cohorts were markedly below those observed in the control and 0.10 % LBP cohorts ($P < 0.05$). SOD activity demonstrated progressive elevation with increasing LBP concentration, with control cohorts exhibiting markedly reduced levels versus all LBP-treated cohorts ($P < 0.05$).

3.5. Liver immune gene expression analysis

The impact of dietary LBP levels (0.00 %, 0.10 %, 0.20 %, and 0.40 %) on hepatic *HSP70*, *AKP*, and *SOD* gene expression in Bulatmai barbels are provided in Fig. 1. As illustrated in Fig. 1, liver *HSP70* gene expression exhibited a gradual decline with elevated dietary LBP supplementation levels, although no statistically significant variations were observed between cohorts. The *AKP* gene expression in the liver decreased gradually with the increase in polysaccharide concentration, and the *AKP* gene expression in the control cohort was markedly higher than that in the other three cohorts ($P < 0.05$); The *SOD* gene expression level in the liver of the control cohort and 0.10 % LBP cohort displaying notably elevated expression versus the liver of the 0.20 % and 0.40 % LBP cohorts ($P < 0.05$). There was no significant difference in *SOD* gene expression in the liver between the control and 0.10 % LBP cohorts.

3.6. Analysis of intestinal flora diversity and richness

Per the species accumulation curve (Fig. 2a), the curve gradually flattened with a gradual increase in sample size, indicating that the sample size in this experiment was adequate, and species richness could also be predicted. The Venn diagram illustrates shared and distinct amplicon sequence variants (ASVs) in each cohort. The four cohorts (i.e., control, 0.10 % LBP, 0.20 % LBP, and 0.40 % LBP) had 235 ASVs and had their own unique ASVs, 293, 272, 96, and 65, respectively (Fig. 2b). These results indicate that LBP could alter the composition of the intestinal flora in Bulatmai barbel. The α diversity was analyzed using Chao1, Shannon, and Simpson indices (Fig. 2c). It was found that adding 0.10 % LBP to the diet could elevate the diversity and richness of intestinal flora in Bulatmai barbel. The α diversity of 0.10 % LBP cohort was higher than that of the control cohort, but it was lower than that of the control cohort with increased LBP concentration. PCoA was used to analyze β diversity, and the diversity of species differences among different cohorts could be seen in Fig. 2d.

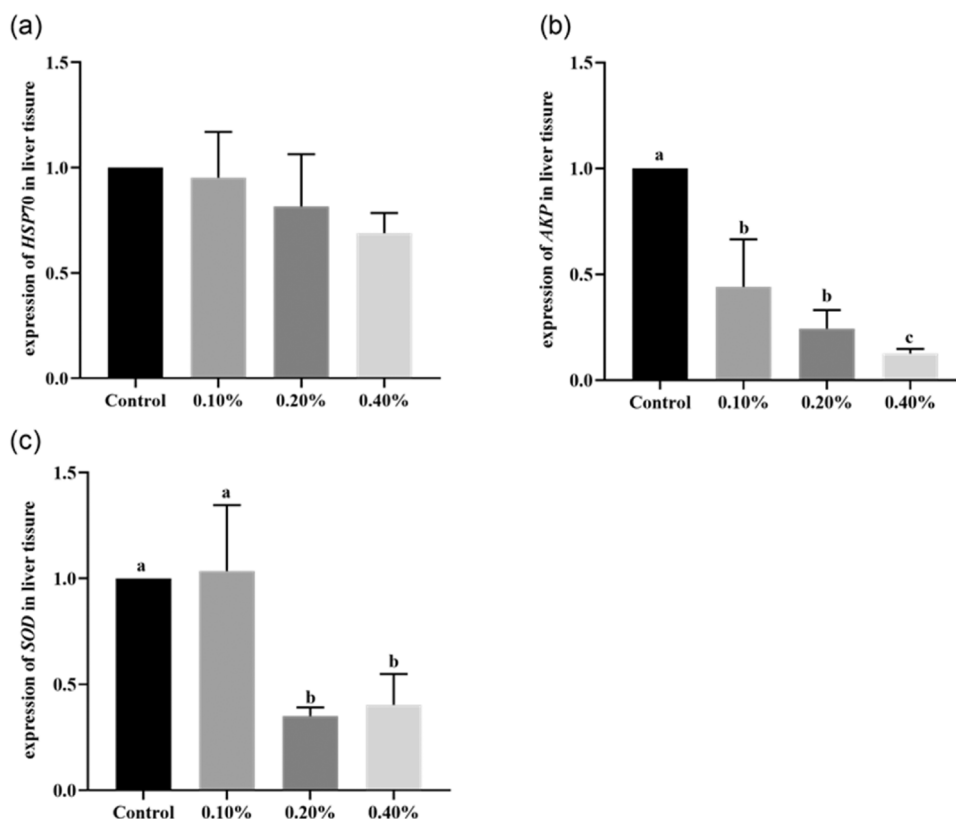


Fig. 1. Effects of dietary LBP levels (0.00 %, 0.10 %, 0.20 %, 0.40 %) on hepatic HSP70, AKP and SOD gene expression of Bulatmai barbel.

3.7. Analysis of intestinal flora composition and structure

We examined the intestinal flora composition at the phylum and genus levels, classified them, and plotted a stack histogram (Fig. 3). *Cetobacterium* was found to be the dominant genus. The relative abundance of *Cetobacterium* was notably elevated in the experimental cohort versus the control cohort. As LBP levels elevated, the proportional presence of *Cetobacterium* showed a corresponding rise (Fig. 3a). At the phylum level, fusobacteria and proteobacteria were the main phyla identified. When compared to controls, fusobacteria showed enhanced proportional presence following LBP administration, and this elevation continued progressively with higher LBP levels, suggesting LBP's supportive role in fusobacteria multiplication and development. Conversely, the controls exhibited higher proteobacteria proportions than LBP-treated samples. The proportional presence of proteobacteria showed a declining trend with elevated LBP concentrations, demonstrating LBP's suppressive effect on proteobacteria growth (Fig. 3b). In addition, the distribution and variability of intestinal flora at the phylum level among the control, 0.10 %, 0.20 %, and 0.40 % LBP cohorts could also be observed through the visual clustering heat map (Fig. 3c).

4. Discussion

Research has demonstrated the beneficial growth effects of LBP. Previous investigations indicate that incorporating suitable quantities of LBP into feed rations considerably enhances weight increment and growth parameters in *Ctenopharyngodon idella* (Mo et al., 2016) and groupers (Tan et al., 2019), leading to enhanced developmental outcomes. Our experimental findings revealed that LBP supplementation yielded varied improvements in Bulatmai barbel's weight gain and growth rates while decreasing feed conversion ratio, hepatosomatic index, and viscerosomatic index. Despite lower feed conversion ratios and hepatosomatic indices in experimental treatments compared to controls, these differences lacked statistical significance. The 0.10 % LBP cohort exhibited notably superior final mass and weight increment compared to controls, whereas both high and low LBP concentrations demonstrated markedly reduced weight gain and specific growth rates. These observations suggest dual effects of LBP, with optimal supplementation levels proving advantageous for Bulatmai barbel's developmental performance.

Serum biochemical indices can be used to evaluate the nutrient uptake, absorption, and metabolism in fish (Li et al., 2024). Serum TP, ALB, and BUN reflect the absorption and conversion efficiency of dietary proteins and the health status of animals. Protein synthesis efficiency in the body increases when the urea nitrogen content is diminished (Chashnidel et al., 2020). GLU is the primary energy source of the body, and changes in its content reflect a dynamic balance between carbohydrate synthesis and metabolism (Borer

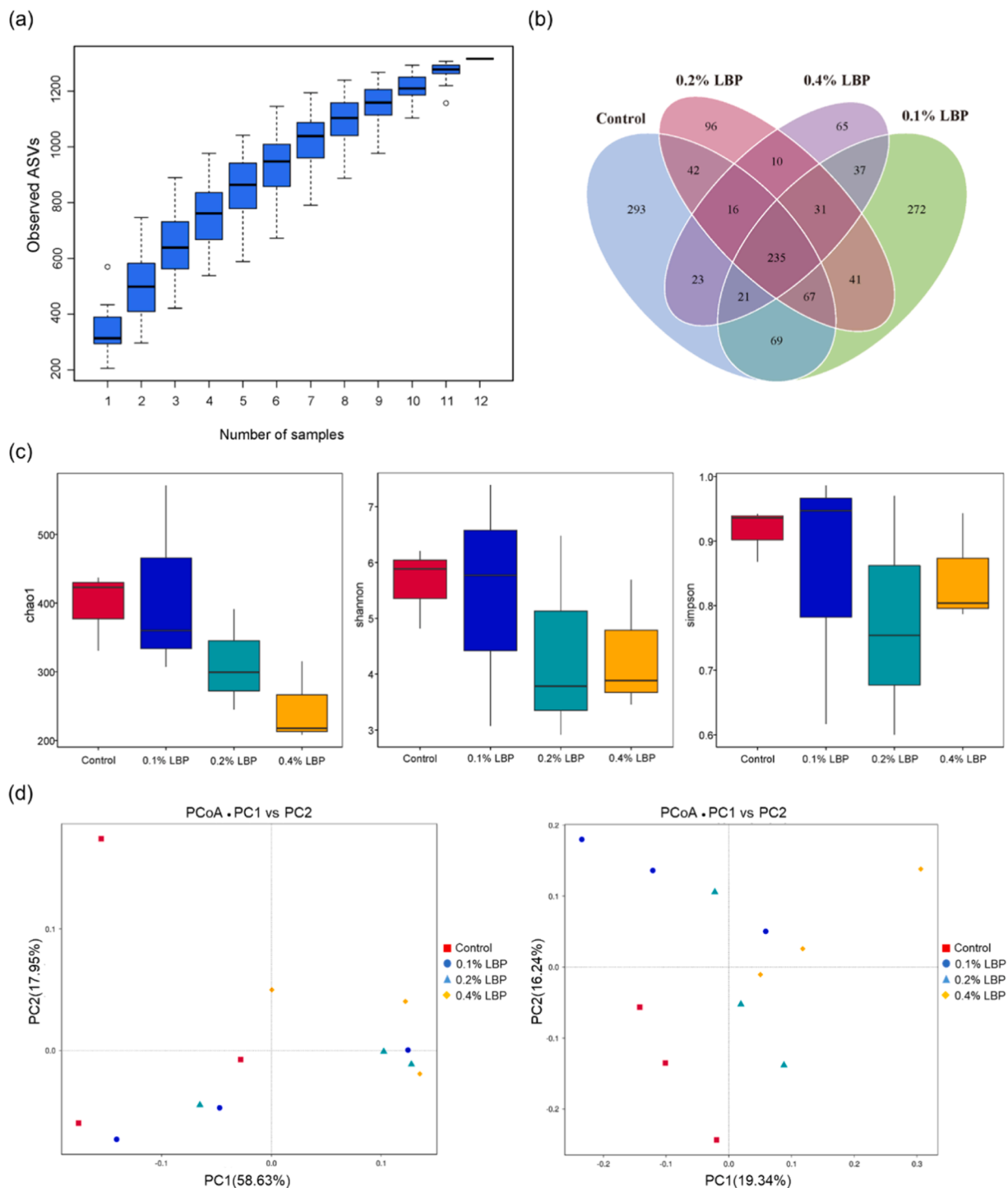


Fig. 2. Effects of dietary LBP levels (0.00 %, 0.10 %, 0.20 %, 0.40 %) on intestinal microbial diversity and richness of Bulatmai barbel. (a). Species accumulation curve; (b). Venn diagram; (c). α diversity index; (d) Principal coordinate Analysis (PCoA).

et al., 2012). Serum TG and CHOL concentrations serve as indicators of lipid metabolic processes. The present investigation revealed markedly reduced TG and BUN concentrations in Bulatmai barbel serum following various LBP supplement treatments versus control cohorts. Although other serum biochemical parameters exhibited variations relative to the control cohort, these differences lacked statistical significance. The 0.10 % LBP supplementation resulted in decreased GLU, CHOL, and TG values compared to controls, demonstrating LBP's sugar and lipid-reducing properties. Reduced BUN values observed in 0.10 %, 0.20 %, and 0.40 % LBP cohorts

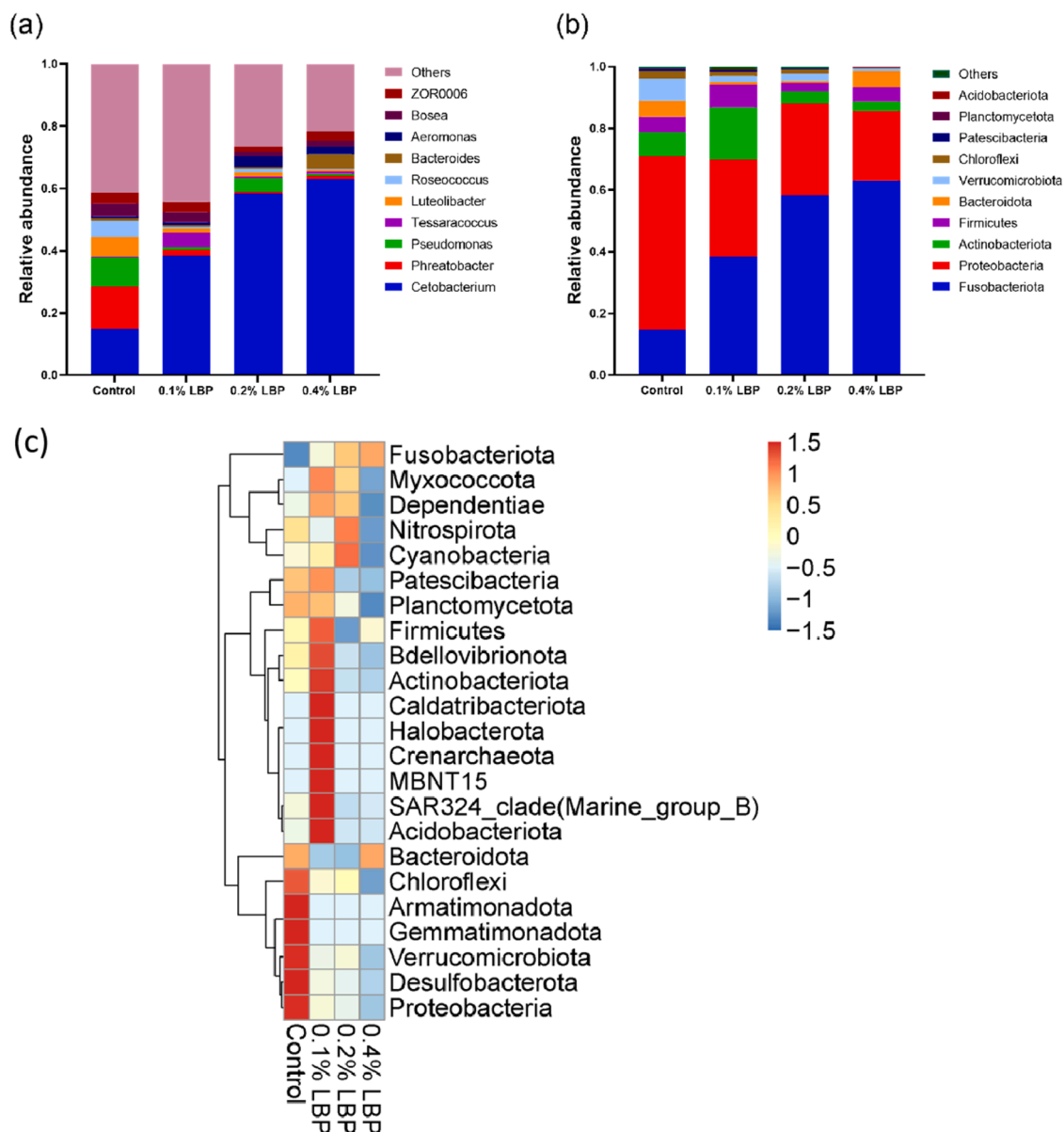


Fig. 3. Effects of dietary LBP levels (0.00 %, 0.10 %, 0.20 %, 0.40 %) on intestinal flora composition and structure of Bulatmai barbel. (a). at the genus level analysis; (b). at the phylum level analysis; (c). Clustering heat maps of relative species abundance at the phylum level.

suggested that appropriate LBP supplementation enhances protein utilization efficiency. Regarding ALT and AST, these enzymes function as physiological damage markers. Under healthy conditions, ALT and AST maintain low concentrations; however, physiological damage triggers abnormal elevations in serum ALT and AST levels (Zhang et al., 2019).

The intestinal tract is an essential digestive tract for organisms, and its digestive ability is an important index affecting healthy growth. The digestive enzyme activity reflects the level of digestion and nutrient absorption in the intestinal tract. The higher the activity of the digestive enzymes, the faster the growth rate of the body. Several studies have shown that polysaccharides improve the activities of intestinal digestive enzymes in fish. Zahran et al. (2014) observed that supplementing with optimal amounts of astragalus polysaccharide notably enhanced intestinal amylase function in *Oreochromis niloticus*. Findings indicated that LBP supplementation improved intestinal enzyme activities in Bulatmai barbel, with trypsin and lipase levels showing marked elevation in 0.10 %, 0.20 %, and 0.40 % LBP cohorts versus controls. Conversely, amylase levels in the 0.20 % and 0.40 % LBP cohorts showed no significant variation from control cohorts. Nevertheless, this observation warrants additional study. Overall, enzymatic activity peaked when LBP supplementation was maintained at 0.10 %.

The liver serves as an essential organ influencing fish weight and generating various immune and antioxidant enzymes. ACP functions as a characteristic lysosomal enzyme that participates in phagocyte intracellular digestion, nucleic acid metabolism, protein

processing, phosphate ester breakdown, immune system regulation, and vital biological processes (Li et al., 2009). As an absorbing enzyme, AKP is distributed in the intestinal tract of fish and is involved in the absorption and transport of lipids, glucose, calcium, inorganic phosphorus, and other nutrients (Tengjaroenkul et al., 2000; Cara et al., 2003). T-AOC, etc are antioxidant enzymes secreted by the liver that affect the body's antioxidant capacity. Polysaccharides can enhance the activities of immune and antioxidant enzymes, thereby improving the immune capacity and antioxidant performance of the body. Liu et al. (2010) found that feeding ginseng polysaccharides markedly improved the antioxidant capacity of prawns versus the control cohort. The findings of this investigation demonstrated that ACP activity notably diminished and AKP activity markedly elevated after the addition of dietary LBP, which requires further investigation. After adding an appropriate amount of LBP, the antioxidant capacity of the liver improved, and the content of MDA decreased markedly, suggesting that the body can remove excessive free radicals to improve antioxidant capacity in the body. Studies have shown that polysaccharides can enhance immune organ development, activate macrophages, improve the phagocytic ability of macrophages (Ren et al., 2014; Li et al., 2020), promote the proliferation of B and T lymphocytes (Zhao et al., 2017; Bo et al., 2018; Wang et al., 2018; Zhang et al., 2019), and induce the activation and maturation of dendritic cells (Zou et al., 2011; Park et al., 2014). The mechanism underlying the antioxidant activity of polysaccharides can be summarized as follows: (1) polysaccharides can directly act on free radicals and remove them (Pu et al., 2016); (2) polysaccharides can improve the activity of antioxidant enzymes (Chai et al., 2018; Lin et al., 2020); and (3) polysaccharides can regulate antioxidant signaling pathways (Yang et al., 2014; Xiao et al., 2018).

Heat shock proteins (HSPs) are highly conserved. This protein is produced to reduce damage to the human body when the external environment stimulates it. HSP70 is one of the most important HSPs family members. AKP is an important indicators of the immune ability of fish. SOD is an important antioxidant enzyme and the primary substance for scavenging free radicals in the bodies of fish. SOD plays an essential role in catalyzing the disproportionation of superoxide into hydrogen peroxide and oxygen (Andreyev et al., 2005). In this study, the introduction of LBP showed that hepatic *HSP70* gene expression remained comparable between treatment and control cohorts, while *AKP* expression exhibited significant downregulation. The *SOD* expression in the 0.10 % LBP cohort maintained levels similar to controls, whereas both 0.20 % and 0.40 % concentrations led to notable decreases in *SOD* expression. Additional research is necessary to elucidate the mechanisms behind the observed *AKP* expression reduction.

As prebiotics, polysaccharides facilitate food digestion and nutrient absorption, enhance antioxidant capacity, and regulate the intestinal microbial environment, thus promoting the healthy growth of the body and maintaining the intestinal ecological balance (Wang et al., 2020; Liu et al., 2022). This investigation examined the impact of LBP feeding on the intestinal microflora in Bulatmai barbel. The intestinal microflora configuration and organization were changed by α diversity and β diversity versus the control cohort, suggesting that LBP can change the intestinal microflora structure in Bulatmai barbel and regulate the intestinal microbial environment. In this investigation, LBP enhanced the growth of fusobacteria and inhibited the growth of proteobacteria. Fusobacteria maintain intestinal environmental homeostasis by producing butyrate (Bennett et al., 1993; Liu et al., 2020). Many investigations have suggested that proteobacteria, a sign of many diseases, can diminish the relative abundance of probiotic bacteria, modulate the intestinal flora, and contribute to the healthy growth of the body (Rizzatti et al., 2017; Liu et al., 2020). At the genus level, LBP increases the relative abundance of *Cetobacterium*. *Cetobacterium* promotes sugar utilization and insulin expression, thus reducing blood sugar levels and regulating overall physical well-being (Xun et al., 2019).

5. Conclusions

In conclusion, a diet supplemented with LBP fed to Bulatmai barbels for 8 weeks markedly improved growth performance, reduced feed coefficient, enhanced digestive capacity, immune capacity, and antioxidant performance, and regulated immune-related gene levels. In addition, LBP can markedly change the composition and structure of the intestinal microflora and promote healthy growth in Bulatmai barbels. This study showed that the optimal supplemental level of LBP in diet is 0.10 %. LBP can be used as a natural immunomodulator in aquaculture, and the mechanism by which LBP affects the intestinal microflora in Bulatmai barbel should be further explored.

CRedit authorship contribution statement

Zhao Zhigang: Funding acquisition, Conceptualization. **Wang Wei:** Validation, Supervision, Resources. **Zhang Rui:** Software, Investigation, Formal analysis. **Wang Shihui:** Software, Investigation, Formal analysis. **Xu Wei:** Software, Resources. **Guo Kun:** Software, Investigation. **Meng Xianwei:** Software, Formal analysis, Data curation. **Luo Liang:** Writing – original draft, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.anifeedsci.2025.116315.

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